

ELI — Memory

ELI v2.4.69

September 2026

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ELI Memory Subsystem

Updated for v2.4.69. Memory is governed by a storage policy (eli/memory/policy.py): every row has an origin, repeats are merged, weight follows a forgetting curve reinforced by use, and faded derived rows are archived. Recall applies a time window before ranking, and reports what it did. Turn retrieval is shared in eli/memory/retrieval.py; FAISS deletes use tombstones.

eli/memory/ holds 8,770 lines in 11 files: relational, full-text, vector and graph storage, all local SQLite and FAISS. Companion to project_overview.md.

The two layers

ELI’s memory is not “a dump of every interaction”. There are two layers:

1. **Conversation history** (conversation_turns, conversations, session_summaries): broad and chronological. A turn is kept unless the persistence gate (runtime/persistence_gate.py) rejects it as junk. This is what a question about a period of time reads.
2. **Distilled memory** (memories, semantic, user_patterns, the knowledge graph): a smaller set that the gate and the storage policy judged worth keeping, labelled by origin, merged when repeated, and faded when nothing reinforces it. Faded derived rows move to memories_archive instead of being deleted.

The memory-runtime report states this explicitly (EXPLAIN_MEMORY_RUNTIME, memory internals), with live counts for each layer.

Files

File	Lines	Role
memory.py	5,677	the Memory class (69 public methods), schema, DBPaths, upkeep, module facade
policy.py	118	pure storage-policy functions (origin, dedupe key, strength, archive, merge)
retrieval.py	209	shared turn retrieval (retrieve_for_turn), turn cache, time window
claims.py	206	dated claims about the user: valid-from and valid-to, learned-at, supersession
unified_retrieval.py	168	the orchestrator stages consume retrieve_for_turn through it; formats the verified-memory block
vector_store.py	637	FAISS index, embedder, tombstones
knowledge_graph.py	643	entity and relation graph
habits_memory_db.py	466	habit and legacy memory helpers
system_index.py	278	indexed apps, executables, files
memory_truth.py	188	read-only inspection used by status surfaces
memory_adapter.py	131	compatibility adapter
__init__.py	278	facade (get_memory, get_agent_memory, path helpers)

The storage policy (policy.py)

Every stored row gets an **origin** from `classify_origin(source, kind, tags, text)`:

Origin	Meaning	Factor on half-life
user_said	the owner said it or confirmed it	1.0
eli_said	ELI's own assistant text	0.5
tool	executor or observation output	0.35
news	news digests and reflections	0.25
telemetry	bookkeeping (is_bookkeeping_memory)	0.15

Rules, all in `policy.py` and pinned by `tests/test_memory_policy.py`:

- **Dedupe.** `text_key` is a sha1 of the normalised text (first 20 hex characters). A repeat bumps `seen_count` and `last_seen` instead of adding a row; `merge_group` keeps the earliest event time.
- **Strength.** $2^{**} (-days_since_last_touch / half_life)$, floored at `MIN_WEIGHT` (0.05). The half-life is the base half-life $\times$ origin factor $\times (1 + 3 \times importance) \times (1 + 0.6 \times \ln(1 + uses))$, where `uses` are repeat sightings plus recalls.
- **Base half-life.** 30 days, scaled by the square root of the median gap between days the owner used ELI, clamped to 14-120 days (`adaptive_half_life_days`).

- **Pinning.** A user_said row with importance ≥ 0.85 stays at strength 1.0.
- **Archiving.** should_archive is true only for a derived (not user_said) row that has never been recalled, has importance < 0.85 , and whose weight is ≤ 0.12 .
- **Vector index.** Telemetry rows are not embedded (wants_vector_index).
- **Evidence about the user.** Only user_said counts (counts_as_evidence_about_user).

Columns added to memories: event_ts, seen_count, last_seen, last_recalled, recall_count, origin, text_key. Tables added: memories_archive, memory_meta. The semantic table gained evidence_count, last_seen, evidence.

What counts as the same fact, and what strengthens a memory

- **Dedupe key.** normalise_text lower-cases, single-spaces and trims sentence punctuation from the edge of each word, and keeps meaningful marks. “I love C++”, “I love C#” and “I love C” are three facts, “balance -5” is not “balance 5”, and “3.5” is not “35”. “Not fine!” and “not fine” are one. A key-version marker makes upkeep recompute old keys once (rekey_text_keys); it never merges.
- **Retrieval is exposure, not use.** Recalling a memory only increments exposure_count. Its strength is reinforced by record_recall_outcome(ids, helped=True), which the engine calls when the next user message does not correct the answer those memories went into; a correction lowers the weight and increments corrected_count. A wrong memory that keeps being retrieved cannot make itself stronger. recall_count now means answers a memory helped with, and only that protects a row from archiving.
- **What enters durable memory.** persistence_gate.should_store_memory_text rejects filler and reactions (“haha”, “lol that is funny”, “hello there”), text under four words, questions, and ELI’s own internal output. Conversation turns are still logged in full; only durable memory is filtered.

Dated claims (claims.py)

memory_claims holds what ELI believes about the user as bitemporal rows: valid_from/valid_to (when it was true) and recorded_at/superseded_at (when ELI knew it). A first-person statement such as “I work nights now”, “I moved to Berlin”, “I work at Acme Corp” or “my dog is called Max” becomes a claim when it is stored. A single-valued relation supersedes the old value and keeps it as history, “I used to work nights” is recorded as history only, “I no longer work nights” retires the standing claim, and the same value again confirms instead of duplicating. Memory.claims_current(), claims_during(start, end), claims_known_at(when) and claim_history(relation) answer “what is true now”, “what held in spring” and “what did you believe on 1 March”. Retrieval adds the relevant claims to the evidence (a period question gets what held then), and deleting the source memory withdraws its claims. Extraction is a small exact set of patterns, not a model call.

Daily upkeep

Memory.run_upkeep() runs at most once a day (upkeep_due()), started asynchronously at engine start and after responses (upkeep_async()), and skipped under ELI_TEST_MODE. Steps:

1. **backfill:** label existing rows with origin, key and event time (after copying the database to a .pre_policy.bak file the first time). A dry run reports what it would fill and writes nothing; 1a. **rekey:** recompute dedupe keys once after the key definition changes;
2. **consolidate:** merge exact duplicates and rebuild memories_fts;
3. **decay:** recompute weight with the policy strength (idempotent: a pure function of age, importance and use, so a missed run cannot leave a row over-weighted);
4. **archive:** move faded derived rows to memories_archive (search_archive, restore_from_archive bring them back);
5. **identity:** merge legacy user ids into one stable owner id (owner_id, kept in memory_meta); only UUID-shaped legacy ids are merged;
6. **learning:** remove repeated identical belief revisions, cap corroboration at the number of days the owner has used ELI, and drop ledger copies of replay rows;
7. **semantic:** fold repeated facts into one row with evidence_count and up to five quotes.

A run only marks the day done when every step succeeded; otherwise `run_upkeep` reports status: partial with the failed steps, and upkeep retries after an hour. `integrity_report()` reports the state of all of this for the health checks: ok is false when vectors are missing, and status is healthy or degraded.

The Memory class (`memory.py`)

A single class that owns most persistent concerns:

- **Memory:** `store_memory`, `add_memory`, `recall_memory`, `search_memory`, `search_memories`, `memories_between`, `get_recent_memories`, `adjust_weight`, `apply_weight_decay`, `consolidate_memories`, `archive_faded`, `search_archive`, `restore_from_archive`, `store_episodic`, `store_semantic`, `recall_semantic`, `store_reflective`.
- **Conversation:** `add_conversation_turn`, `store_conversation`, `get_conversation_history`, `get_recent_conversations` (with `since`, `until`, `role`), `get_recent_turns_since`, `search_conversations`, `get_turns_for_day`, `save_session_summary`, `get_session_summaries`.
- **Habits:** `log_habit_event`, `get_habit_events`, `add_habit_rule`, `get_habit_rules`, `get_detected_habits`, `record_habit_run`.
- **Self-improvement and learning:** `log_learning_event`, `log_failure`, `log_correction`, `add_observation`, `log_improvement`, `add_capability_proposal`, `propose_capability`, `get_pending_proposals`, `get_recent_failures`, `get_recent_improvements`, `get_recent_observations`. These tables are written on the **agent** database (`SelfImprovementEngine` opens `get_agent_memory()`); the user database holds empty copies.
- **Health:** `get_stats`, `get_dashboard_counts`, `get_db_routing_info`, `integrity_report`.

`vector_store` is a lazy property; the knowledge graph is reached through `knowledge_graph.get_knowledge_graph()`.

Schema

The blank `user.sqlite3` template holds 27 tables and three FTS5 indexes. `memories` (+ `memories_fts`), `memories_archive`, `memory_meta`, `semantic`, `conversation_turns`, `conversations`, `session_summaries`, `kg_entities` (+ FTS), `kg_relations`, `recall_log`, `runtime_events`, `learning_replay`, `observations`, `habits`, `habit_events`, `habit_rules`, `user_patterns`, `user_model`, `eli_stances`, `belief_revisions`, `corrections`, `failures`, `error_tracking`, `improvements`, `capability_proposals`, `news_articles` (+ FTS), `news_reflections`. `agent.sqlite3` holds `agent_dispatches`, `agent_metrics` and the self-improvement tables.

`emotion_events` is owned by `cognition/emotion_timeline.py`, not the Memory class: it opens `user.sqlite3` directly and creates its table on first use. Columns: `ts`, `user_id`, `session_id`, `detected`, `expressed`, `valence` (negative, positive or neutral), `confidence`, `source`, `arousal`, `user_text` and `eli_prior_action` (the action ELI ran on the preceding turn, so it can ask whether the mood turned because of something it did). Reads come back ORDER BY `ts` DESC, `id` DESC; the `id` tiebreak matters because several reads can land within one second.

Each conversation turn is stored in `conversation_turns`. A `learning_replay` row is also written per turn (it carries action, outcome and reward), and the evidence ledger records the turn once in `runtime_events`; replay rows are not mirrored into the ledger.

Shared turn retrieval (`retrieval.py`)

The single owner of semantic and conversation recall for a turn. `BusMemoryAgent` and the orchestrator both call `retrieve_for_turn()`, so a query is not searched twice with different budgets.

- **Turn cache:** 8 seconds per process. The key includes session, user, query, time window and every limit (`semantic_limit`, `conv_limit`, `recent_limit`, `summary_limit`, `hop2_limit`, `merge_cap`, `verified_only`), so a deeper pass never gets the shallower result. `invalidate_turn_cache()` clears it.
- **Time window first.** When the planner finds a period in the question (`query_planner.parse_window`), memories dated inside it are fetched by date (`Memory.memories_between`, ordered by importance ×

weight) and merged with the topic hits, hits outside the window are dropped, and the user's turns inside the window are read from `conversation_turns`. The result carries `window_stats` (candidates, added by date, in window, turns) and the orchestrator logs and reports it (`memory_diag`).

- **Explicit dates.** `parse_window` reads ISO dates ("2026-03-03"), "3 March", "March 3rd 2025", ranges ("between 1 May and 10 May"), "in June 2025" and "during 2024" as well as relative periods. A date with no year is its latest past occurrence.
- **Token budget.** `context_budget.chars_per_token()` measures the loaded model's own tokenizer and falls back to 3.5 characters per token.
- **Hop-2 deepening** when the first hits are sparse; **heuristic rerank** via `rerank_candidates()`; **contradiction detection** through the bus.
- **Dated evidence.** Each recalled line carries its event date (`evidence_format`); working memory keeps the original date rather than the date it was re-pinned.

recall_memory: the hybrid retriever

The primitive `retrieve_for_turn()` builds on:

1. **FAISS and FTS5 both run** and are fused by reciprocal rank (`fuse_ranked_lists`); a LIKE scan covers the case where FTS returns nothing. `keyword_only=True` skips FAISS.
2. **Noise filtering** (`memory_exclusion_sql` in `core/self_provenance.py`): ELI's own bookkeeping kinds and sources never resurface as recalled user memories.
3. **Importance-weighted ordering** via `COALESCE(importance, 0.5)`.
4. **Reinforcement.** A recalled row gets `last_recalled` and `recall_count` updated and its weight reset to 1.0; the recall is written to `recall_log` with its `memory_id`.

Column detection (`_memory_table_columns`) guards against schema drift between versions.

Vector store (vector_store.py)

FAISS `IndexFlat`, embeddings from a local nomic embedder (`llama.cpp`).

- `_embed_lock` (`RLock`) serialises embedding: the embedder is not thread-safe, which is why orchestrator retrieval is sequential.
- Metadata lives in `meta.json` (migrated from a legacy `meta.pkl`).
- Singleton via `get_vector_store()`; shutdown-aware; `reset_vector_store()` for rebuilds.
- **Tombstones:** `mark_memory_deleted(id)` writes to `meta.tombstones.json` next to the metadata; search skips tombstoned rows without a rebuild; `compact_tombstones(live_ids)` reclaims space.
- `store_memory()` returns `vector_indexed: bool`, a failed index write logs at warning level, and `fire_memory_uncertainty_event()` drives the World tab's `memory_uncertainty` bar.
- Tombstoning is a per-call-site discipline: there is no shared `delete_memory_row()` helper, so any direct `DELETE FROM memories` must tombstone the vector itself (`consolidate_memories` and `profile_extractor._scrub_onboarding_snapshot` do).

Knowledge graph (knowledge_graph.py)

`kg_entities(name, type, aliases, description, confidence)` and `kg_relations(subject_id, predicate, object_id, weight, source)`: a subject-predicate-object graph. FTS5 over entities with insert, update and delete triggers for fuzzy search_entities. `upsert_entity`; `context_for_prompt` gives lightweight SQLite-only prompt context with no embedding. A stop-word list stops common words becoming entities.

Truth layer (memory_truth.py)

`inspect_sqlite` and `inspect_vector_store`: read-only inspection used by status surfaces; reads `vectors/meta.json` and falls back to the legacy pickle. Backs `truth_report` and the `memory-status` surfaces.

Promotion across tiers

Promotion is a fan-out from `user_patterns`:

```
turns → user_patterns ─┬→ semantic (profile_extractor._promote_to_semantic)
                        └→ KG entities and relations
                           (persona_updater._populate_kg_from_user_patterns)
```

`_promote_to_semantic` applies a durable/transient prefix filter, dedupes against semantic and fires only when `_insert_user_pattern` actually inserted, so reaffirmations do not re-promote. Corroboration counts days the owner reaffirmed a fact, not extraction passes, and a passing mention cannot overwrite a fact the owner stated and reaffirmed (`_insert_user_pattern` weighs provenance and corroboration; each supersession is recorded in `belief_revisions`, and repeats are deduplicated). `store_episodic` and `store_semantic` are thin aliases and are not the promotion path.

Measuring it

`tools/eval/memory_bench.py` runs 11 deterministic scenarios against a temporary database, with no model: extraction, multi-session recall, a relative and an explicit time window, knowledge update, abstention, provenance, deletion, no destructive merge, no self-reinforcement, and chatter not becoming durable knowledge. They follow the LongMemEval ability categories but are written for ELI; they are not the official dataset. `tests/test_memory_bench.py` runs them, so a memory change that regresses one fails the suite.

Honest assessment

- **Strong:** genuinely hybrid (vector, FTS5, graph) on one local SQLite foundation; a measurable storage policy with an audit trail; recall that filters by date before ranking; noise filtering that keeps ELI's own reflections out of recall; embedder serialisation that avoids the segfault.
- **Weak:**
 1. `memory.py` is a 5.7k-line class spanning unrelated concerns (semantic, conversation, habits, learning, failures, capabilities, upkeep). It wants splitting along those seams.
 2. **Schema sprawl.** memories and a legacy memory, conversations and `conversation_turns`, plus a standalone semantic table; inline column detection compensates for schema churn.
 3. **Narrow intake.** Promotion works, but it is fed by regex extraction plus one LLM pass per session, so `user_patterns`, the semantic tier and the knowledge graph stay small relative to the conversation history. Widening the extractor is the work.
 4. **Retrieval cannot detect nonsense.** Similarity bands for genuine and gibberish queries overlap, so no absolute similarity floor separates them; the relative cutoff in `vector_store` tightens the candidate pool but is not a relevance oracle.
 5. **No shared delete-and-tombstone helper** (above).

Recent-history window

The user-facing tunable `cog.mem_recent_turns` (Settings ► Cognition; default 24, maximum 80) controls how many recent turns enter the prompt. `RECENT_HISTORY_CAP` no longer exists (the old memory-evidence module was removed); the prompt assembler and `context_budget` do the budgeting, and a recall question keeps a floor of a third of the window for memory context.